Maximum Velocity of Human Movement used for Human Fall Detection using a Support Vector Machine classifier for Doppler Radar Sensors Data

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***Abstract*— The feasibility of detecting falls among different human activities based on micro-Doppler signatures is investigated. This study uses the University of Glasgow Radar Signature dataset, consisting of recordings from 72 persons performing six different activities, including falls. The maximum velocity that each person reached while performing an activity was extracted from the Doppler spectrogram and used to train a Support Vector, to distinguish falls from other activities. The model's performance is evaluated, to assess the potential of using maximum speed as the classification feature on real-world applications for fall detection.**

Keywords—Fall detection, Support Vector Machines.

I. Introduction

"Population ageing is a triumph of humanity but also a challenge to society" [1]

Approximately 28-35% of people aged of 65 and over, fall each year. This figure increases to 32-42% for people over 70 years of age [2]. The situation is not different in Mexico. According to a document published by CENAPRECE [3] 65% of older adults ("adultos mayores" in the original document in Spanish) who don't live alone, suffer falls. 40% of those who live on an assisted living facility and 20% of those who are interned in a hospital fall at least once a year. Accidents, both inside and outside the home, are the 6th leading cause of morbidity in people over 65 and are the 5th leading cause of death in older adults. Of these accidents, 70% are due to falls.

Old people will inevitably fall with a certain frequency: according to [2] and [3], once a year on average. Even if a person never leaves home, or even if the person lives in an assisted living facility, falls may occur. They may occur when an older person is alone, in the bathroom, in the kitchen, in the garden. Falls may be caused by poor vision, a slippery floor, unstable furniture, even a pet. Nooruddin et al [5] have pointed out that making the entire living environment fall-proof is not a feasible solution.

According to an article by Blackburn et al [6], after a person has fallen, and is unable to stand up by himself, the main risk derives from the "long lie", a term for the physical and clinical outcomes in older people who lie a substantial amount of time on the floor before receiving help. The same article also states that even when a personal alarm was available, when they have fallen, the older persons infrequently used the alarm to request help. The article states that a better way to avoid this "long lie" problem is to use smart technology for automatic fall detection to automatically notify caregivers and emergency services about the accident.

Around the world, many organizations and researchers are approaching the problem from three different points of view: 1) preventing falls, 2) getting help for a person when a fall has just occurred and 3) giving the proper medical care and rehabilitation after the fall. This study focuses on technologies for automatic fall detection, which addresses item 2: getting help when a person has fallen and can't get up by himself.

There are many options for fall detection. On their 2014 article, Delahoz and Labrador [7] present a classification scheme for Fall Detection Systems (FDS):

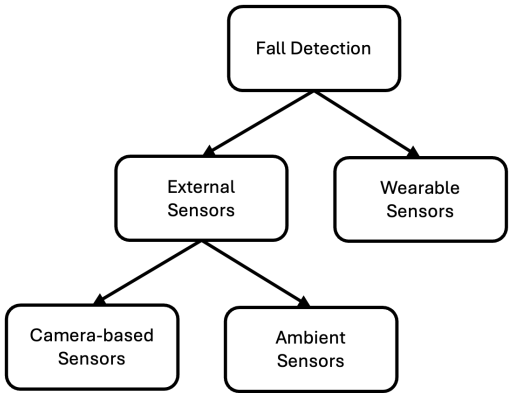


Fig. 1. Fall detection systems classification. (*Delahoz and Labrador [7] p.19821 Fig. 6*)

According to the previously mentioned work by Blackburn et al, fall detection by external sensors should be preferred over wearable sensors because they do not impose any burden on the person being protected by the system.

The goal of this work is to evaluate the use of maximum speed detected by a Doppler Radar, to classify a human action as a fall. The use of just one feature for classification may be useful for applications that need to respond on real time. The work uses a Support Vector Machine to classify the signals.

II. Related Work

Nooruddin et al [5] reviewed many FDS, spanning a very comprehensive spectrum of technologies, including single-sensor and multi-sensor systems, and wearable and external systems. They evaluated the systems using the Accuracy, Sensitivity and Specificity metrics typically used in fall detection literature to rate these systems. They concluded that the number of sensors does not normally impact the system`s accuracy. They also noted that accuracy is more dependent on the features selected for classification in the system. Their article identifies strengths and weaknesses for the different sensor technologies: accelerometers are the best for wearable solutions; Kinect sensors cannot be used outdoors because sunlight affects their accuracy and their ability to estimate depth. Sound sensors are ubiquitous and nonintrusive but can generate false alarms due to ambient noise and raise privacy concerns. Wearable motion sensors are compact and respect privacy, but they require frequent charging. Computer-vision or surveillance-based fall detection systems need huge amounts of processing power. Hsu et al [8] state that all those issues can be avoided with millimeter-wave radar or microwave sensors, using electromagnetic waves Doppler effect. Hu, Cao et all explore the details of Wi-fi and radar sensors [9]: they are noninvasive, they ensure privacy. Radio wave technology can detect through obstructions like walls, is not affected by environmental conditions such as poor lighting, and can operate efficiently in low power. Another important factor for FDS, is cost, since for FDS be practical, they should be designed to be both very reliable and affordable, so they are widely available for people who need them.

In the past few years, deep learning and related classification technologies have gained considerable interest in radar-based FDS, because they extract salient features from radar signals in an automatic fashion. The problem is that deep learning methods require a large amount of training data, and the actual fall detection requires a large amount of computational and memory resources. Most current deep learning approaches require a long latency even at the inference/testing stage, because of complex data processing methods or deep neural networks. Current research is very focused on accuracy, ignoring computational cost. Li, Le Kernec et al [10] have explored the requirements of FDS systems to be deployed on resource-constrained platforms. The consideration of the balance between efficiency and performance is a main item when selecting the most suitable algorithms. A natural step, after classification algorithms development is how to approach a real-time implementation of radar-based fall detection in constrained platforms.

III. Method

A. Software tools

MATLAB R2025b is used to process the data and to test the different data processing methods to transform the input signal, extract the relevant features and identify falls. The steps of the method to train the model and to classify the data are documented as MATLAB code. Different MATLAB libraries were used to process and transform radar data, for training the model and for testing it. The MATLAB code is useful for rapid developing and testing different models and approaches, and it is also useful to write production quality software programs for the training component and for the classifier component.

B. Training and test data

The University of Glasgow Radar Signature dataset [11] was used. The data gathering process is documented at the URL shown in reference [11]. In summary, data was generated by a Frequency Modulated Continuous Wave (FMCW) radar, operating at the 5.8 GHz band, with a chirp (a frequency sweep) that was repeated each millisecond and to digitize it, was sampled 128 times per sweep. Note that the term “beat” is sometimes used instead of “sweep” in FDS articles. The signal has a 400 MHz bandwidth. The data contains a record for each sample, that is a complex number, where the In-phase data for the sample is the real part and the Quadrature data for the sample is the complex part. Each file contains either 10 seconds or 5 seconds of radar data for a motion capture, i.e. 1,280,000 records or 640,000 records. Also, each file includes four records; the first one contains the band (5.8 GHz), the second has the chirp duration (1 ms), the third has the number of samples per sweep (128) and the fourth shows the bandwidth (400 MHz).

There is a total of 1,754 files, recorded by 72 participants aged between 21 and 98 years old. The total space required by the data files is 15.9 Gigabytes. Each participant performed six activities in front of the radar device. The radar has one antenna for sending the signals and one for receiving the signals. The person does not use any sensors. Older individuals didn’t perform the “Falling” activity, to avoid the possibility of injuries for them. Because of this, it can be seen on Table I that there are less files for activity number 6: Falling. Each activity was repeated 3 times. The activities were performed indoors. Only a single person is measured each time. The following table summarizes the data files information:

TABLE I. Glasgow Data Files

Activity number	Description	Number of files	Activity duration (seconds)	Size (Mbytes)
1	Walking back and forth	312	10	15.4
2	Sitting down on a chair	312	5	7.7
3	Standing up from a chair	311	5	7.7
4	Picking up an object	311	5	7.7
5	Drinking water	310	5	7.7
6	Falling	198	5	7.7

File names have a structure that gives information about its contents. The first character of the name is the Activity Number from 1 to 6 (First column of Table I). Character 2 is a ‘P’ and characters 3 and 4 are a number from 01 to 72, identifying the person who executes the activity: for example, person ‘P01’ is person number 1. Character 5 is an ‘A’ and characters 6 and 7 are a number from 01 to 06, stating the Activity Number again. Character 8 is letter ‘R’ and character 9 is a number in the range 1-3; this indicates which repetition of the activity is recorded in the file. The file extension is ‘.dat’. For example, file

1P01A01R2.dat has the data for Person 1 executing the second repetition of activity 'Walking back and forth'.

A sample of the files was used for training the model. Since the files record either a fall or an activity that is not a fall, the variable we wish to predict is a binary variable, and a binomial distribution is appropriate for this kind of data.

As seen on Table 1, there proportion of files for each activity is 17.7% of the total file count, except for Fall files. The proportion is:

$$p = \frac{198}{1754} = 0.1129$$

The formula for calculating the sample size for a dichotomous (binary) variable [12] is based on the statistic properties of the binomial distribution:

$$n = \frac{Z^2 \cdot p \cdot (1-p)}{d^2}$$

Where:

- Z is the critical value for the desired confidence level
- p is the expected proportion of falls on the data
- d is the desired margin of error

Selecting a 95% confidence level, we have $Z = 1.96$, $p = 0.1129$, $d = 0.05$, we have n = 153.8819, so the sample size will be 154 files.

For training the model, files were selected randomly for each activity, according to the following table:

TABLE II. FILES SELECTED FOR TRAINING THE MODEL

Activity number	Description	Number of files
1	Walking back and forth	28
2	Sitting down on a chair	28
3	Standing up from a chair	27
4	Picking up an object	27
5	Drinking water	27
6	Falling	17
Total		154

C. Classifier for Fall Detection

Hu et al [9] surveyed 74 studies on radar-based fall detection. The survey includes different radar formats, different classifiers, and the rating of each study's model using Accuracy, Precision and Recall. Given the goal of detecting falls with technologies that require low resources, compared to those used typically by Deep Learning solutions.

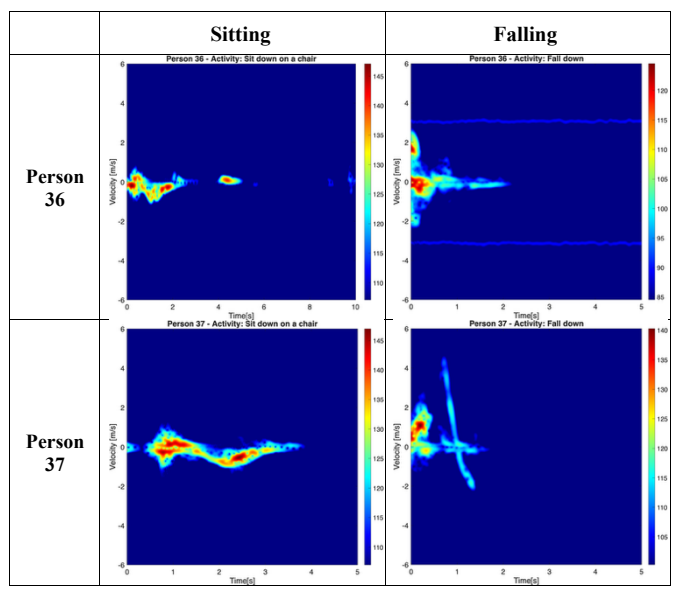
Support Vector Machines (SVM) has been used extensively for its superior performance over other classification methods, such as the Fisher Linear Discriminator and the Bayesian Method (Kim et al [13]). Many Radar researchers have applied SVMs successfully for array signal processing, On Table 1 of [9], 20 studies used Convolutional Neural Networks (CNN) as a classifier, while 19 studies used SVM. The third most-used classifier is K-Nearest Neighbors (KNN) with 12 studies. As shown on Tables 2-5 of [9], SVM has a very good performance, compared to CNN.

MATLAB supports SVM training and prediction on their "Statistics and Machine Learning Toolbox".

The SVM will be fed an interval of Doppler signals gathered by a Radar device and returned as a series of samples for an interval of time, and from them, it will classify if the recorded activity is a fall or not.

D. Feature Selection and Extraction.

Data files contain a record for each one of the 128 samples taken for each "chirp" echoed by the target, i.e. the person who is performing the activity in front of the radar. A chirp is sent every millisecond, and the activity is recorded for 5 seconds, except for the "Walking back and forth" activity which lasts 10 seconds: 5 for walking forth and 5 for walking back. The first step is to rearrange the data in a matrix where we have one row for each millisecond, and each row has 128 columns, with the samples. The next step is to take the Fast Fourier Transform (FFT) for the matrix, using a Hamming window to reduce the noise. This transforms the matrix to Velocity vs Time. If we apply a second FFT, we obtain the Doppler matrix, the spectrogram, which characterizes each action. Let's look at the spectrograms for Persons 36, 37 and 38, during two activities: sitting on a chair, and falling:



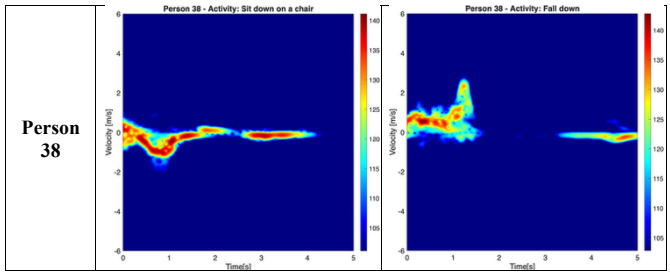


Fig. 2. Spectrogram comparison of two activities for three different persons.

Figure 2 clearly shows that spectrograms for the same activity for different persons have a similar signature, while spectrograms of different activities have clearly different signatures, so velocity is a feature for identifying falls.

When a person falls, the sudden change of position manifests as an increase in velocity. The University of Glasgow data has 5-second samples, so the feature will be the maximum velocity recorded during the 5 second interval. Because of this, maximum velocity in the interval will be used for the SVM. The following block diagram shows how the data is processed and classified.

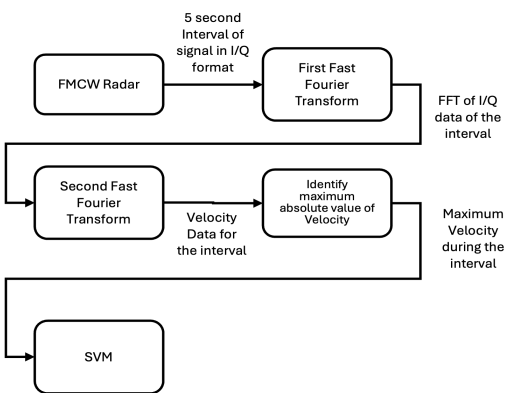


Fig. 3. Data processing blok diagram

E. *Training.*

MATLAB was used to train a SVM model. Function 'fitscsvm' was used specifying a linear kernel function. Three-hundred sixteen files were selected at random from the files, to be used for training, and the rest of them were used to evaluate the model.

F. *SVM Model evaluation.*

Hu et al [9] use Recall, Precision and F1 to evaluate the SVM models in their survey, so those metrics will be computed for the proposed model and compared against those shown on Table 3 of [9] to determine the quality of the model. Shalev-Schwartz and Ben-David [15] explain how to calculate the metrics.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \quad (3)$$

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (4)$$

$$F1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}} \quad (4)$$

IV. Results

The SVM processed the rest of the University of Glasgow files. with 154 files: see Table II.

There were a total 72 participants recorded by the radar, so the training sample has records for most of the participants. The number of samples for falls is low, and the cause of this is that it was difficult for the University of Glasgow to record many persons falling, and because of the possible risk, none of the older persons performed any falls. Also, there are close to three repetitions for each participant for each activity, so this is a representative sample.

After training the SVM with this data, another sample with 316 files was obtained to forecast falls. The following table shows the summary information.

TABLE III. Sample for testing the SVM

	Count	**Repetitions**		
Persons in the sample	70	**1**	**2**	**3**
Walking back and forth	48	16	13	19
Sitting down on a chair	59	22	23	14
Standing up from a chair	60	17	16	27
Picking up an object	55	15	25	15
Drinking water	57	15	23	19
Falling	37	12	13	12

The testing data is also a representative sample of the University of Glasgow data files.

The following table shows the metrics obtained for the testing data and compares them to results reported on Hu et al [9].

TABLE IV. Comparison between SVM models

		Metrics		
Article	**Motions**	**Precision**	**Recall**	**F1**
This article	Six activities, 37 falls	75%	72%	75%
Garripoli et al. [16]	Four activities, 65 falls	100%	100%	100%
Amin et al. [17]	19 types of falls, 14 types of nonfalls	100%	82%	Not available
Hanifi et al. [18]	543 falls and 675 nonfalls	88%	87%	88%
Erol et al. [19]	Four activities, 106 samples	97.7%	94.7%	96.2%
Erol et al. [20]	Four activities, 106 samples	99.6%	99.6%	100%

		Metrics		
Erol et al. [21]	25 falls, 47 nonfalls	100%	95%	97.5%

V. Discussion

The metrics obtained for this SVM are not good enough to be used for a real application. There may be two reasons: either the feature is not adequate, or the model is not adequate. To eliminate the possibility that the problem lies on SVM, further work is needed: 1) Establish a baseline, using Logistic Regression. 2) Train two other models, k-NN and Random Forest. The comparison of the results of the SVM against these other models should allow to evaluate if the problem lies with the usage of SVM. To eliminate the possibility that the problem lies on the selected feature, a pair of new features should be used: Standard Deviation of the Velocity and Acceleration look promising. The state of the art of fall detection obtains higher metrics than what was achieved on this work, so this further work is desirable. Using a set of more sophisticated features than those used on this study may also be done. For example, in [20] combined Range and Doppler features and obtained very high metrics.

VI. Conclusions

A better model should be built, trying different classifiers and different features. A better model should be useful to establish a baseline for developing a system that can be used in real world applications, and this is desirable, because the costs of current radar-based configurations are prohibitive for massive deployments at hospitals and assisted care facilities, and for domestic use.

Li, Le Kernec et al [9] reported in 2023 a very exciting technique where they used a magnitude thresholding approach, which allows to pinpoint the region where the activities features are located, separating them from the signal's noise. Also, this technique reduces the amount of data to be processed, reducing both the training time and memory footprint.

VII. Limitations and Future Work

This model was not designed to be able to run on a continuous basis: it assumed that data is provided in files that record 5-second intervals. A workable solution should be able to process data on a continuous basis. A promising project is to use Li and Le Kernec's techniques [9] and determine how the model should be built such as to be able to process continuous signals. The use of boosting algorithms like XGBoost or LightGBM may help to solve the problem of processing data on a continuous basis.

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